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A structural and palaeomagnetic study of a section through the eastern Subbetic, Southern Spain

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Abstract: A well-exposed 2 km section through the Eastern Subbetic, immediately to the south of the Crevillente wrench fault zone can be divided into an upper thrust sheet of Triassic rocks, a middle thrust sheet of Cenomanian–Turonian basinal sediments, and a lower thrust sheet of Jurassic limestones unconformably overlain by Upper Cretaceous–Palaeocene sediments. Structural relationships suggest that the middle sheet overthrust the lower sheet; and both units were subsequently folded and imbricated. This sequence was then overthrust by the upper sheet, incorporating a footwall duplex of Cenomanian–Turonian sediments. These structures can be related to the complex deformation associated with a single overthrusting event. The dominant thrust transport direction is to the east, strongly oblique to the overall trend of the orogen. However, a palaeomagnetic study isolates stable components which indicate no cumulative vertical axis rotation of the thrust sheets. The section is cut by strike-slip faults whose displacements appear to be relatively minor. The Triassic rocks of the upper thrust sheet form a highly disrupted thrust sheet extending over much of the Subbetic, rather than being local to the Crevillente fault zone. This section through the Subbetic can be interpreted as the result of a major east-directed thrust of basinal sediments of the Median Subbetic over marginal shelf facies of the Internal Subbetic, strongly oblique to the general ESE trend of the orogen. The shortened sequence is cut by a later set of strike slip faults which are probably related to movement on the Crevillente fault.

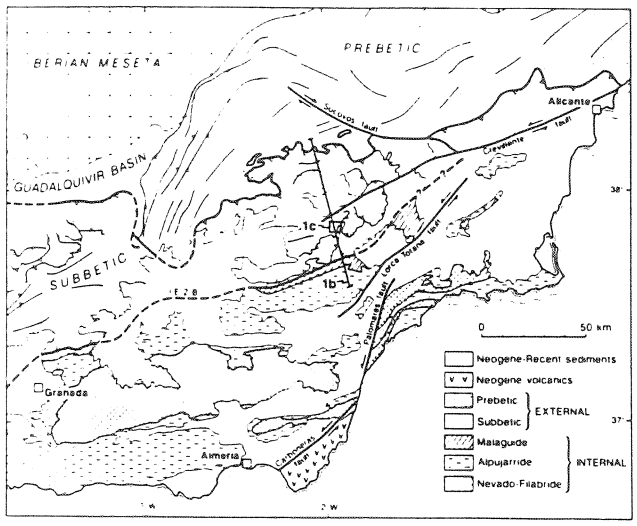
The Betic Cordillera of Southern Spain are an example of an orogen resulting from highly oblique convergence, showing evidence for both compressional (e.g. Egeler & Simon 1969) and strike-slip (e.g., De Smet 1984; LeBlanc & Olivier 1984) tectonics, and for significant clockwise rotations (e.g. Osete *et al.* 1989; Platzman 1992). Following an initial compressional phase, the orogen has undergone extensional collapse (Platt & Vissers 1989), generating a series of graben in the Internal Zones and the Alboran sea basin to the south. Key to the understanding of this orogen are the relative timing of the various tectonic episodes, and the relative transport directions. The latter objective can be difficult to determine in the presence of extensive vertical-axis rotations.

In this contribution we report palaeomagnetic and kinematic data from a critical section covering a major back-thrust within the Subbetic Zone, immediately to the south of the Crevillente strike-slip fault which is considered by De Smett (1984) to be the major controlling feature of Subbetic deformation. Palaeomagnetic data (documented in this paper) demonstrate that the footwall of this thrust has not rotated by a significant amount, in contrast to much of the surrounding region. Kinematic data from this regional scale thrust can thus be related to the orogen as a whole. This study provides constraints on the relative timing and transport directions of the major structures of the central Subbetic Zone, and allows us to develop a discussion of the tectonic evolution of the frontal part of the Betic orogenic wedge, involving a change from oblique thrusting to strike-slip deformation.

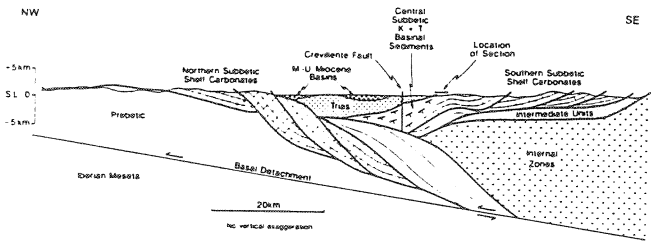
Structural context

The External zones of the Betic Cordillera form a fold and thrust belt developed on the southern continental margin of Iberia in Miocene times (Fig. 1 a & b). They lie north of the Internal or Betic zone, which has experienced a much longer history of deformation and metamorphism commencing in Late Cretaceous or Palaeogene time (Egeler 1964; Torres-Roldan 1979; Bakker *et al.* 1989). Deformation in the External Zones of the Eastern Betic Cordillera began in Burdigalian (early Miocene) to Langhian (mid-Miocene) time (Hermes 1978), but the nature of this deformation has been the subject of debate. Many workers have considered the region to be a classic foreland fold-and-thrust belt, with the dominant sense of tectonic transport to the north (Egeler & Simon 1969; Paquet 1969; Garcia-Hernandez *et al.* 1980) or northwest (Banks & Warburton 1991). Others have noted the importance of south-directed thrusting along the southern margin of the Subbetic (Macgillavray 1964), and still others the effects of dextral wrench faulting (Hermes 1978; Paquet 1972; De Smet 1984; LeBlanc & Olivier 1984) particularly in the Late Miocene–Recent. De Smet, in particular, considered the whole Subbetic Zone to have been formed by distributed dextral shear centred on the Crevillente fault zone during the Miocene. LeBlanc & Olivier (1984), following Andrieux *et al.* (1971), suggested that deformation in the External Zones was primarily a consequence of the westward motion of an Alboran microplate (comprising the present Alboran Sea and the Internal Zones of the Betic and Rif mountain belts) relative to both Iberia and Africa.

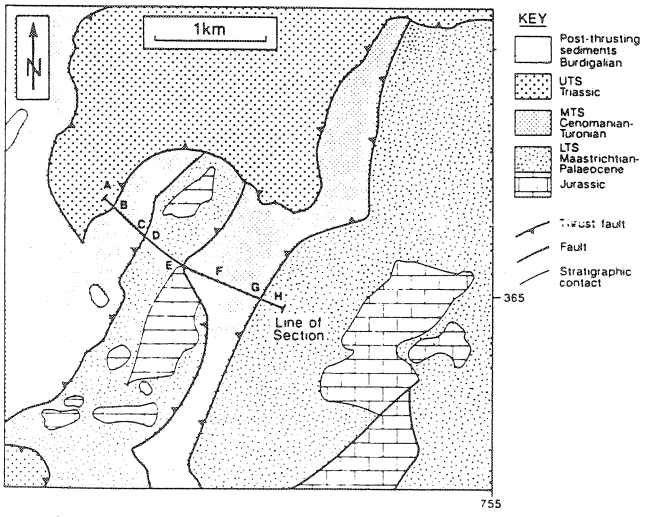
Given the variety of models for the tectonics of the Subbetic, the published kinematic data which may be used to test them is sparse. Frizon de Lamotte *et al.* (1989) have published a regional kinematic study of the Internal Zones which indicates a general westward sense of displacement and Lonergan *et al.* (1994) produced a more detailed study of the kinematics of the Internal External Zone Boundary in the Eastern Betics. Recent palaeomagnetic



(a)



(b)



(c)

Fig. 1. Location maps and structural section. (a) Geological map of SE Spain showing the location of the study area (c) and of the structural cross-section though the Subbetic shown in (b). IEZB marks Internal–External Zone Boundary. (b) Structural cross section through the Subbetic of the Eastern Betic Cordillera, modified after Banks and Warburton (1991) and Allerton *et al.* (1993). Location shown on (a). The detailed structural section that is the subject of this paper is located on this section. (c) Geological sketch-map showing location of the structural section (largely based on IGME sheet 931, 1974). Grid from IGME sheet.

studies in the External Zones of the Betic Cordillera have shown that there is a consistent pattern of clockwise rotations about vertical axes (Osete *et al.* 1989; Platzman 1992). This is clearly a significant result for kinematic studies, as kinematic data can only be related to any external framework if the amount, timing and mechanism of vertical axis rotations are understood.

The stratigraphy of the Eastern Subbetic has been the subject of study of a group from Amsterdam, synthesised by Hermes (1978). The oldest rocks encountered in the Subbetic are Triassic continental and marine facies. Overlying these are a sequence of Lower Liassic carbonate platform sediments. Late Liassic–Late Jurassic times saw regional differentiation associated with extension and basin formation within the Subbetic; the Northern Subbetic contains condensed pelagic limestones; the Median Subbetic is dominated by marly facies, and in the Southern Subbetic oolitic limestones and other shelf facies have been deposited. The Cretaceous–Burdigalian sequences throughout the Subbetic are dominated by pelagic turbiditic sedimentation. A fuller description of the stratigraphy of the central part of the Eastern Subbetic has been given by Seyfried (1978), who indicates that in this area the Jurassic stratigraphy is highly variable.

Marls and bioclastic limestones of Upper Burdigalian age are unconformable on the rocks that have been involved in the thrusting. They were directly deposited on Triassic, Cretaceous, and Lower Tertiary rocks, and have been gently folded, suggesting that they were deposited during the latest stages of the folding. The overthrust of the Subbetic over the Prebetic cuts Langhian sediments, suggesting that at least this thrust was active until this time.

The Crevillente fault zone which runs approximately ENE–WSW within the Subbetic consists of several individual strike-slip fault strands occuring over a zone about 2 km wide. It cuts Messinian (Upper Miocene) and earlier sediments, and structural evidence suggests that it has exhibited both dextral and sinistral senses of slip during the Neogene.

This study involves a detailed structural and palaeomagnetic analysis of a well-exposed 2 km section along the Baranco Salada river section, trending ESE–WNW through the Subbetic, about 5 km to the south of Crevillente fault zone. The section is at the contact between the Central Zone described by Hermes (1978) as a major wrench zone, and the Southern Subbetic. The aim of the study is to provide well constrained kinematic data which can, with other such data, test models for the tectonic evolution of the Eastern Subbetics.

Stratigraphy of the Baranco Salada section

The structure of the Baranco Salada section is relatively complex, and it has been necessary to construct an informal stratigraphy. We have divided the section into subsections of consistent internal stratigraphy, labelled, from WNW to ESE, from A to H. The stratigraphy of these subsections is indicated in Fig. 2, and the location of the subsections on Fig. 3. From these subsections, three main tectonic units can be identified, each with a distinctive stratigraphy.

Upper thrust sheet

This unit is composed of highly tectonized rocks, which apparently include all the elements of the Triassic sequence

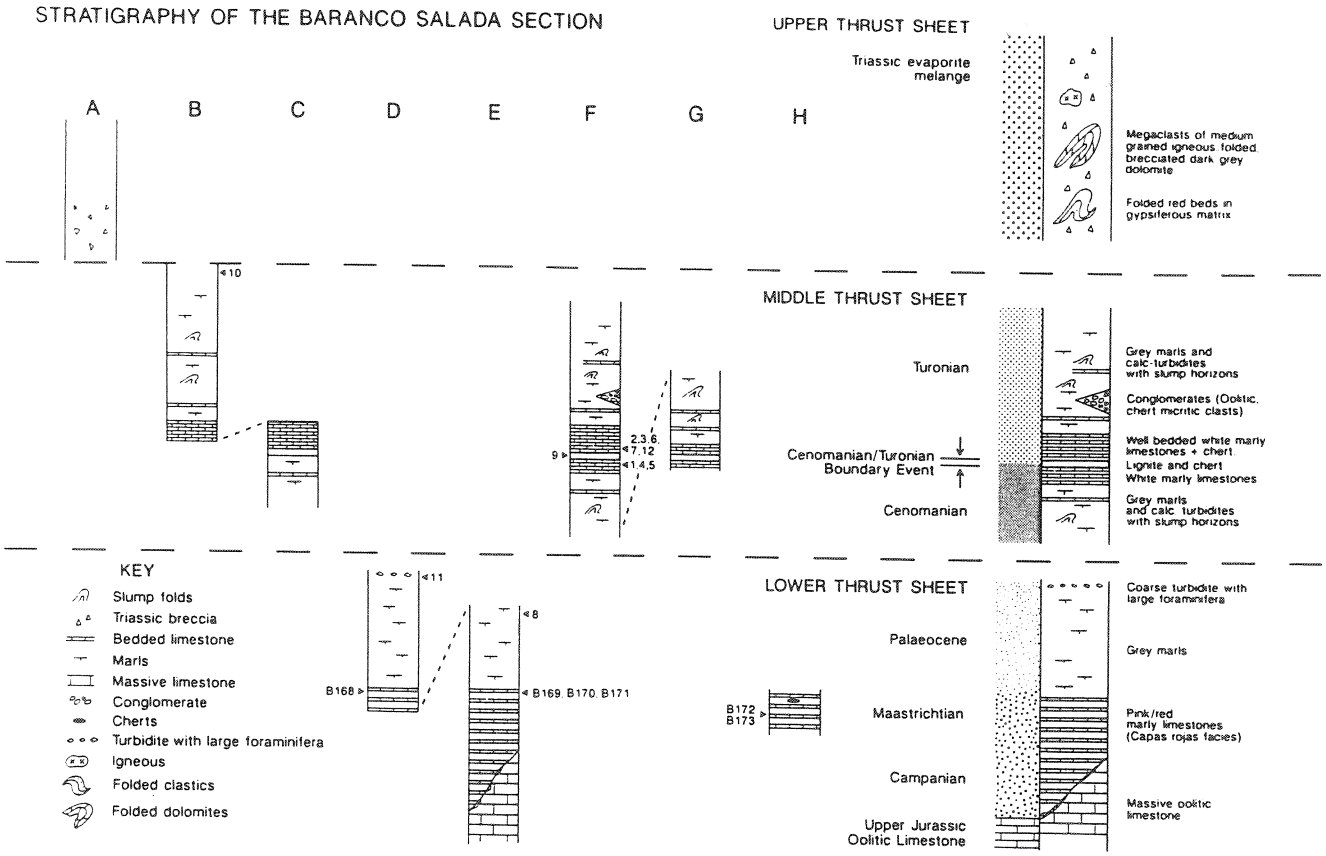


Fig. 2. Stratigraphy of subsections, showing the location of samples for micropalaeontology (1–12) and palaeomagnetic sites (B168–B173). Locations of the subsections are indicated on Fig. 3.

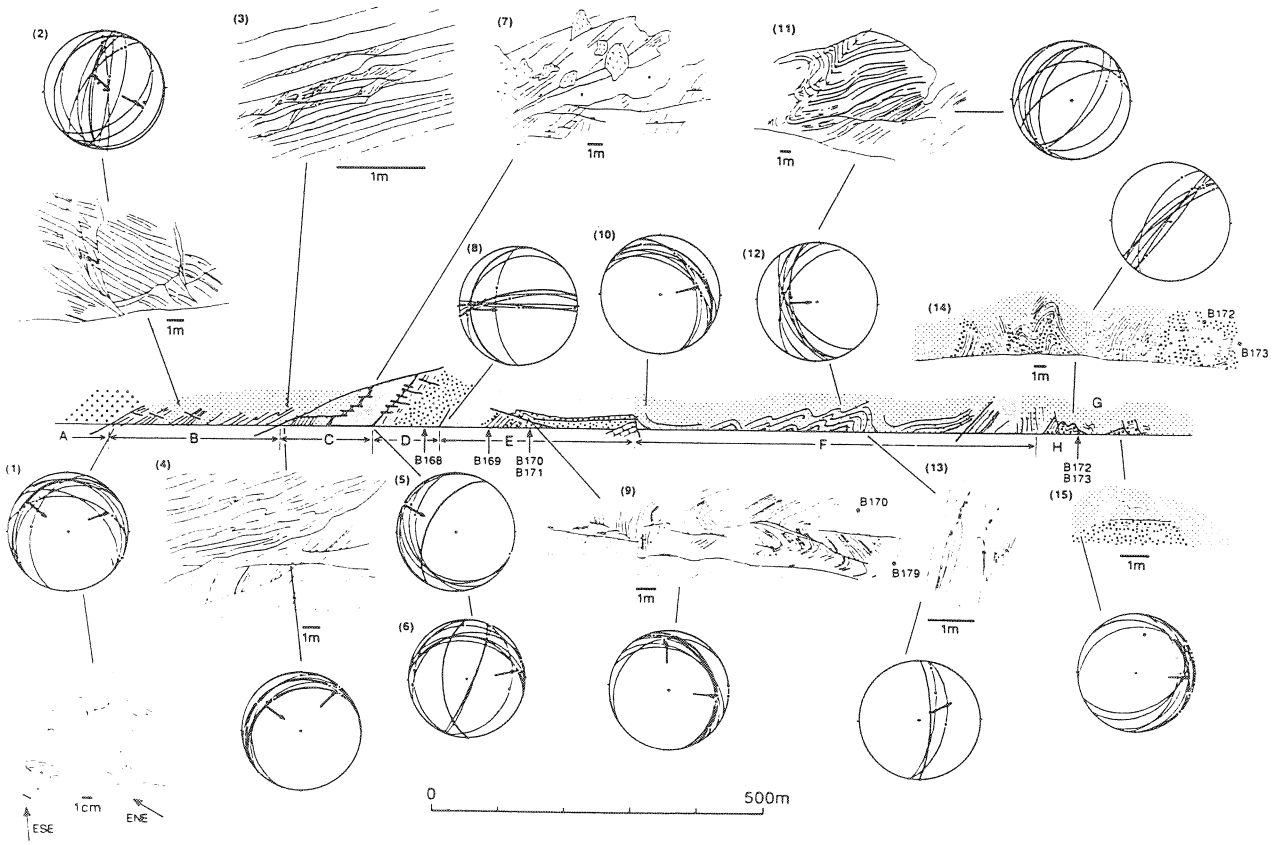


Fig. 3. Structural data from the Barranco Salada section. Kinematic data are illustrated as lineations on shear surfaces on lower hemisphere equal area stereonets. Sense of slip is indicated by direction of transport of the hanging wall. Key to stratigraphy is given in Fig. 2.

in the Subbetic (see for instance, Paquet 1969). Variegated sandstones, dolomites, gypsum and igneous rocks occur as rafts or clasts within a brecciated and recrystallized gypsiferous matrix. Individual blocks are at least metres across, and may extend to kilometres in dimension (e.g. Guadaperos, IGME sheet 931; 751,367)). The thickness of this unit is difficult to establish, as it is so tectonized, and because its upper surface cannot be identified. Elsewhere in the Eastern Subbetic, the Triassic is <950 m thick (Paquet 1969).

Middle thrust sheet

At the base of this thrust sheet is a sequence of interbedded white calc-arenites and light grey pelites. The calc-arenites are frequently coarse grained, with abundant dark organic material. Evidence for slumping and soft sediment deformation is common, particularly in the finer, pelitic layers.

Above this is approximately 3.5 m of dark brown–black organic-rich pelites with black cherts, equivalent to the Cenomanian–Turonian boundary event described elsewhere (Thurow & Kuhnt 1986), which acts as an important stratigraphic marker for this section.

This is followed by at least 10 m of well-bedded, white–light grey micritic limestones with cherts, with dark, organic fragments at their bases. There is abundant evidence for soft-sediment deformation, with the development of isoclinal folds in limestones and cherts.

The uppermost part of the middle thrust sheet is a sequence of interbedded white calc-arenites and light grey pelites. This unit is relatively chaotic, with evidence for extensive slumping. Some turbiditic calc-arenites are isolated as isoclinal slump folds with pinched-out limbs within the grey pelites. Within this part of the section a 3 m thick conglomerate is encountered (see discussion below).

This middle thrust sheet corresponds to the Cenomanian–Turonian sequence of white marls with black and red cherts described in IGME sheet 931 (1974). The top of the middle thrust sheet (immediately below the contact with the Triassic of the upper sheet) is within the Turonian, but the base of the sequence within this section has not been dated. The thickness is uncertain as it is difficult to reconstruct the stratigraphy with any accuracy because of tectonic repetition. It is greater than 50–60 m, although in the majority of outcrops it is slightly less.

Lower thrust sheet

The oldest unit encountered within the lower thrust sheet is a massive oolitic and pisolitic limestone of Late Jurassic age (IGME sheet 931 1974).

The top of the Upper Jurassic oolitic limestones is at a low angle to the bedding in the overlying red marls. Locally, along the contact, brown siliceous material replaces the cement in the limestones, and small, pebble-sized clasts with a grey micritic matrix fill irregularities in the surface. This contact represents an unconformity between the oolitic limestones and the overlying red marls, although there may have been a limited amount of slip along it.

The overlying sequence of thinly bedded red and white marls and marly limestones (capas rojas facies) is dated (IGME sheet 931, 1974) to be from Coniacian to Maastrichtian in age. Foraminifera further constrain the age of the capas rojas in subsections D and E to the Upper Maastrichtian, and those from subsection G to a slightly older, Maastrichtian/Campanian age. The red marly limestones of the capas rojas facies grade upwards into a series of dark grey marls with local turbiditic calc-arenitic horizons. At the top of the sequence these grey marls are overlain by a coarse calc-turbidite bed with abundant large foraminifera. The marls immediately beneath this unit have been dated as Palaeocene.

Turonian conglomerates

A conglomerate horizon with clasts of oolitic limestones, reduced ammonitico rosso facies limestones and cherts is encountered within the Turonian marls and marly limestones of the middle thrust sheet of subsection F. Such horizons have also been described elsewhere within the Subbetic north of Velez Blanco within Upper Jurassic, Berriasian–Barremian, Aptian–Albian, Upper Cretaceous and Tertiary sediments (Kuhry 1975). Kuhry (1975) ascribes these to a source within the Subbetic Upper Jurassic oolitic limestones around Almirez, and argues that these must result from the sub-aerial erosion of an emergent high. Stratigraphic relationships within the Subbetic, and locally within the Sierra de Almirez suggest that this may be part of the source area of these oolitic conglomerates: in subsection E of this section Jurassic oolitic limestones are unconformably overlain by capas rojas facies of Campanian–Maastrichtian age; in a quarry to the northeast of the Piedra de Almirez (Allerton *et al.* 1993) is a spectacular exposure of a bimict angular conglomerate with clasts of ammonitico rosso-facies pink nodular limestones (Tithonian) and Upper Jurassic oolitic limestones, in a micritic matrix containing Oligocene microfaunas. This conglomerate is deposited directly upon Upper Jurassic oolitic limestones, from which they are presumably derived. These relationships suggest that the Cenomanian–Turonian basinal sediments of the middle thrust sheet were relatively proximal to the shelf sediments of the lower sheet.

Structure of the Baranco Salada section

We will first describe the structures observed in the Baranco Salada section from WNW to ESE. We will then discuss the significance of these observations and develop a model to explain them. The data are illustrated in Fig. 3.

Structural observations

The contact between subsections A and B is a thrust fault dipping 24° to the west. Above the contact the Triassic breccia of the upper thrust sheet has a fabric sub-parallel to the contact, marked by imbrication of the clasts and a flow fabric within the gypsum matrix. Beneath the contact, a gouge/Riedel fabric is developed in the grey pelites of the middle thrust sheet. Calcite mineral growth fibres define two sets of lineations on the Riedel surfaces (see Fig. 3 (1)). A clear, early set consistent with the gouge fabric, indicating thrust transport to the ESE is cut and overlain by a later set, indicating reactivation towards the ENE. The gouge/Riedel fabric continues, with variable intensity throughout the upper part of this subsection.

In a series of calc-arenites within this subsection are a set

of conjugate normal faults which have been reactivated as strike-slip faults (Fig. 3 (2)).

Within the well-bedded limestones are a set of small (1 m amplitude), close-tight asymmetric folds, with a variable vergence sense. In addition, there is abundant evidence for bedding parallel slip, with the development of gouge along some bedding surfaces, indicating thrust transport to the SE (Fig. 3 (3)).

The contact of subsection B over C is a shallow angle fault sub-parallel to the bedding of the upper unit, which dips gently to the NW (15–20°). This thrust truncates more steeply dipping beds in the footwall (subsection C, Fig. 3 (4)). The footwall is cut by a set of sub-horizontal faults, with an apparent offset sense of top to the SE. These faults can be interpreted as Riedels to the main thrust. These Riedels have two sets of lineations developed on their surfaces; an early set with a SE sense of shear and a later set, cutting the earlier set, with a NW shear sense (Fig. 3 (4)).

The contact between the Cenomanian marls and limestones (Subsection C) and the Palaeocene marls (Subsection D) is exposed both in the main river and in a small tributary to the SW. In the main river the contact dips at about 50° to WNW, and is cut by a gouge/Riedel fabric with an ESE shear sense (Fig. 3 (5)). This shear fabric is probably associated with the main overthrust contact between subsections B and C, described above, rather than to original slip between the subsections C and D. In the small baranco, the contact dips at about 55° to the west. The gouge/Riedel fabric in the hanging wall preserves a dominant eastward shear sense on gently dipping fractures (Fig. 3 (6)). Shears in the footwall, are steep, parallel to the contact, with associated dextral, strike-slip lineations. In the upper slopes of the main baranco, the overthrust of subsection B truncates the contact between subsections C and D, and abundant Riedels are developed beneath this major fault (Fig. 3 (7)).

The contact between the capas rojas marly limestones (subsection D) and the Palaeocene marls (subsection E), exposed within the baranco, is composite; the upper part dipping 70° and a lower part dipping at 24° NW. Kinematic data suggest that the upper part moved as a sinistral strike-slip fault, and the lower part as a thrust towards the east (Fig. 3 (8)). These may represent two separate phases of motion, or motion in the same sense along an irregular fault surface.

Within the subsection E is a NW-vergent recumbent syncline and anticline with a wavelength of about 3 m (Fig. 3 (9)). A divergent fracture cleavage is developed close to the axis of the fold. Along the lower limb of the syncline is a fault that dips gently (10°) north. Kinematic data from this fault shows at least two movement senses; to the east and to the north. The former is likely to have produced the observed approximately 10m apparent offset of the capas rojas/grey marl contact.

The unconformity between the capas rojas facies marly limestones and the oolitic limestones is exposed on the hill about 300 m NE of the Baranco. In the NW, this contact is gentle, and the capas rojas limestones dip gently NW. In the SE, the contact dips steeply to the S–SE as do the overlying capas rojas limestones. This can be interpreted as part of SE-vergent anticline (although other options such as a drape-fold of capas rojas over a normal fault block are equally possible).

The nature of the contact between the capas rojas and the oolitic limestones in the Baranco is obscure, and the oolitic limestone outcrops only on the S side of the baranco. On its east side it is truncated by a steep sinistral strike-slip faults, striking NW, which juxtapose marly limestones of the middle thrust sheet (subsection F) against oolitic limestones in the south of the Baranco, and with capas rojas facies limestones on the north side. Although the fault making the contact is a clear strike-slip structure, the marly limestone adjacent to the contact preserves a shallow-dipping gouge/Riedel fabric indicative of E–ENE directed shear (Fig. 3 (10)).

The dominant structure throughout subsection F is a set of SE-vergent asymmetric folds (Fig. 3 (11)). These folds are moderately inclined, sub-horizontal, with a slight NE mean plunge, and a wavelength of between about 10 m and 50 m. The folds commonly show thinning on the long limbs, often by shallow angle normal faulting and thickening of the short, steeply dipping limbs. Thrusts within this part of the sequence dip towards the west, and indicate transport towards the east (e.g. Fig. 3 (12)).

Small normal faults are common, often intimately associated with small reverse faults. Both sets show some reactivation as strike-slip faults.

There is some evidence for an early stage of thrusting, from a conjugate thrust pair observed in the steeply dipping limbs of the open folds (Fig. 3 (13)).

The contact between subsections F and G is not exposed. It may be a normal fault, or the long limb of an asymmetric fold, as it appears to bring up a structurally lower level to the surface.

The ESE-most part of the section shows a complex relationship between the capas rojas of subsection H and the white limestones and pelites of subsection G (Fig. 3 (14)). In one locality, a thin sheet of highly deformed, fissile pink marl of the capas rojas facies (between 10 cm and 1 m thick) is intercallated with well-bedded white limestones with cherts. The white limestone sequence includes small isoclinal folds; presumably the result of soft sediment deformation, black cherts, and black pelites and cherts of the Cenomanian–Turonian Boundary Event unit. This

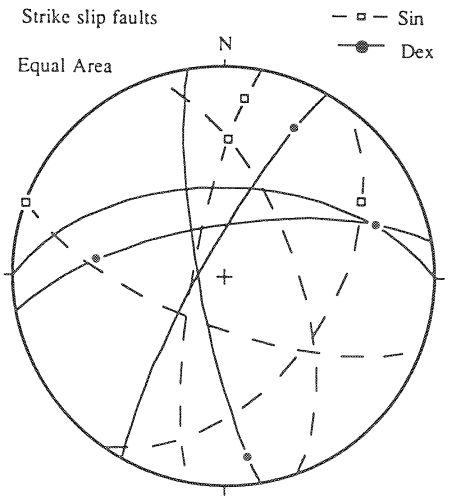


Fig. 4. Orientation of minor strike slip faults in this section. Great circles represent fault planes, and points represent mean lineation directions. Equal area lower hemisphere projection.

repeated sequence is folded in a series of upright, symmetric open folds, with a wavelength varying between about 2 m and 10 m. In turn, these folds have been truncated by steeply dipping strike-slip faults, both sinistral and dextral.

In the farthest SE outcrop of this section we see exposed a shallow contact, sub-parallel to bedding, of capas rojas facies beneath grey pelites of the Cenomanian–Turonian sequence (Fig. 3 (15)). This contact has an early eastward shear sense and a later reactivation towards the NE, consistent with a later set of steeper NE-striking sinistral faults, which cut the shallow-angle structure.

The sequence is cut by a number of strike-slip faults with a range of orientations (see Fig. 4). No consistent pattern has emerged from their orientations; nor do they appear to have significant displacement.

Interpretation

We consider the following observations to be particularly significant.

There is a clear, stratigraphically constrained superposition of Cenomanian–Turonian sediments over Palaeocene/Upper Cretaceous sediments on a fault surface at a low angle to the bedding between subsections C and D, and subsections G and H. The capas rojas sediments appear to be unconformable on Jurassic oolitic limestones and there is no evidence for capas rojas sediments overlying Cenomanian/Turonian sediments. This we take as evidence for overthrusting of middle thrust sheet sediments over the lower thrust sheet.

The thrust contact between the middle and lower thrust sheet itself is tightly folded in subsections G and H and cut by later overthrusts (i.e. B over C/D), so it is a relatively early feature. There is a stratigraphic repetition between subsections D and E, suggesting tectonic imbrication. The steepening of the beds in subsections C, D and E, suggests a duplex structure. As the contact between subsection C and D is also tilted, this imbrication must post-date middle thrust sheet overthrusting.

The faults at the base of subsection B are sub-parallel to bedding and cut steeper beds of subsections C and D (and their contact). This is apparently an out-of-sequence thrust which truncates the culmination of the duplex.

The Triassic of subsection A is thrust over the subsection B. Map relationships (IGME sheets 931 & 932; 1974) suggest that this upper thrust sheet represents a major regional overthrust structure. There are similarlities between the kinematics at the boundary of subsections A and B, within subsections B and at the base of subsections B, which suggest that subsection B may have, for part of its history, been transported within the hanging wall of the upper thrust sheet. It may have originated as a footwall horse. Subsection B appears anomalously thick, particularly within the the well-bedded limestones. This may either be an indication of a thicker, perhaps more basinal original sequence, or of thickening by internal, small scale repetition, for which there is some evidence (see Fig. 3 (3)).

Development of thrust sequence

We offer a relatively simple, though clearly non-unique solution for the origin of structures observed in this section (Fig. 5). This solution relates the structures to deformation in the footwall of a major overthrust which brings basinal Cretaceous sediments (middle thrust sheet) and Triassic sequence of the upper sheet over the more marginal sediments of the lower sheet. We favour this model because it relates the observations to complexities associated with a single overthrusting event, rather than requiring multiple episodes of nappe transport.

The first stage involved thrusting of basinal sediments of the middle thrust sheet over the lower sheet (Fig. 6, a, b). Palaeocene sediments are absent from the lower thrust sheet in the east part of the section, with the middle thrust sheet sequence of subsection G directly overlying capas rojas sediments of subsection H. This apparent cut-off suggests a relative E–W transport sense for the middle sheet over the lower sheet, but this is contradicted by kinematic data from this contact. We believe that this apparent cut-out is either a stratigraphic pinch-out, or the result of a gently-dipping thrust propagating through marginal sediments which dip more steeply to the west.

At some stage, motion on the upper part of this fault ceased, and displacement shifted to a steeper splay lower on the thrust (Fig. 6 b, c). The main thrust may step backward in an out-of-sequence manner, rather than propagating forward through the wedge, because the basinal sediments of the middle thrust sheet are much weaker than the massive shelf limestones of the lower sheet. Further displacement on this splay allowed Triassic (upper thrust sheet) rocks in the lower hanging wall to override rocks of the middle sheet originally from the upper part of the hanging wall, now

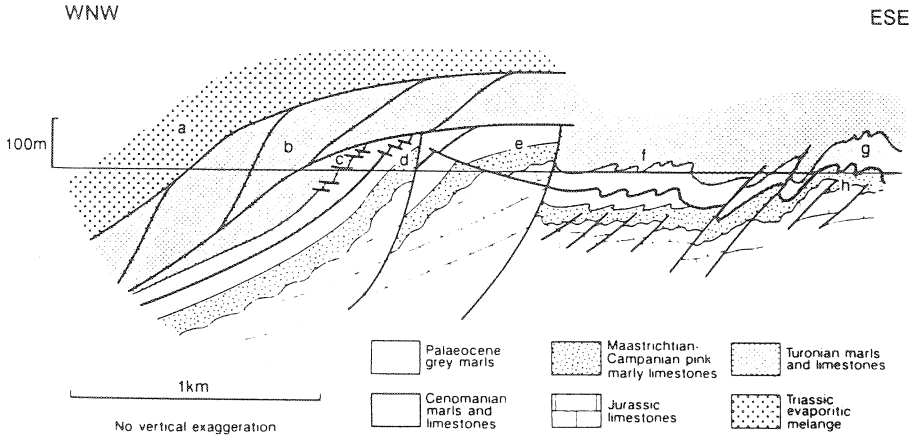


Fig. 5. Interpretative structural cross-section through the Baranco Salada sequence. Key to stratigraphy is given in Fig. 2.

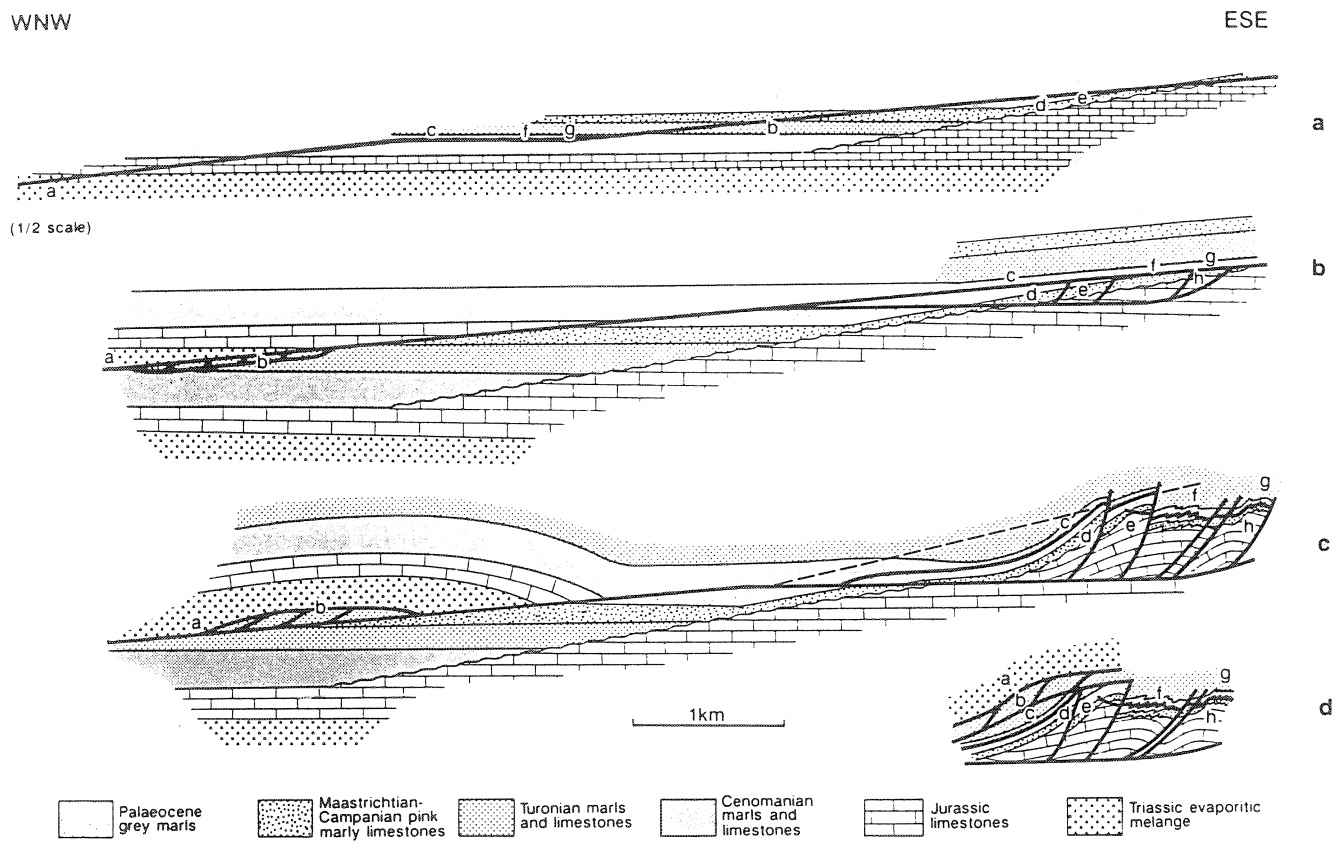


Fig. 6. Tectonic evolution of the Baranco Salada section. Key to stratigraphy is given in Fig. 2.

incorporated in the footwall (Fig. 6 c, d). Continued deformation in the footwall produced the folding and minor thrusting observed in subsections F, G and H, and the formation of the footwall duplex seen in C, D and E. The ubiquitous gouge–Riedel fabric probably formed as a result of continued footwall deformation. Subsection B formed as a footwall duplex that was encorporated with the Triassic sequence of the upper thrust sheet in the hanging wall, and which truncated the culmination of the duplex seen in subsections C, D and E.

The west-vergent fold in subsection E occurs on the culmination of the duplex structure, and may have formed in some 'back-thrust' style relationship to the larger structure. It is truncated by a fault which has evidence for multiple phases of movement. This fault probably initiated as a back-thrust, and reactivated as a normal Riedel to the main overthrust.

Although there is abundant evidence for relatively minor strike-slip, no major faults have been identified, and the dominant part of the section can be explained in terms of thrusting, without needing to evoke major strike-slip displacements. Many minor faults with strike-slip kinematics are observed, but their displacements are thought to be small, as they seldom juxtapose different stratigraphic units. Many of the thrust faults and normal faults in the sequence show evidence for reactivation with a later set indicating transport to the N or NE (e.g. Fig. 3 1,4,9).

Palaeomagnetism of the Baranco Salada section

Some previous work in the Eastern Subbetic has demonstrated large and variable rotations about vertical axes (Allerton *et al.* 1993) which may have occured as a result of Neogene deformation. Obviously, in a study of the kinematics of Neogene deformation, it is critical to know the vertical axis rotation that has affected this region. The Upper Cretaceous capas rojas red marls offer a reliable palaeomagnetic carrier, as demonstrated elsewhere in the Subbetic (e.g. Platzman & Lowrie 1992).

Six sites have been sampled from this facies from the lower thrust sheet, in 3 localities, shown in Fig. 3, and have included two fold tests. Eight 2.5 cm minicores were taken from each site, and typically one or two specimens from each sample. Sample magnetizations were measured on the CCL magnetometer in Oxford, and typical treatment procedure involved stepwise thermal demagnetization until no measurable signal remained. Results were analysed on orthogonal plots using a least-squares routine.

Results typically indicate the presence of two magnetic components (Fig. 7). The first, lower temperature component typically unblocks below 400 °C. It is generally directed north and down, close to the present axial dipole field, and probably represents a recent overprint. The higher temperature component typically has unblocking temperatures between 400 °C and <600 °C. This component is

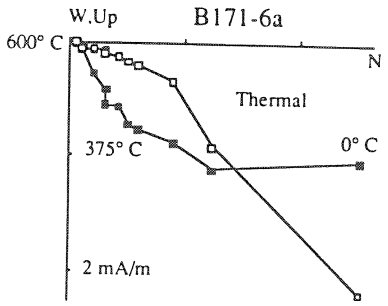


Fig. 7. Typical stepwise thermal demagnetisation behaviour for Upper Cretaceous capas rojas facies samples: orthogonal vector diagrams (*in situ*; open symbols represent points in vertical plane; closed symbols, points in horizontal plane).

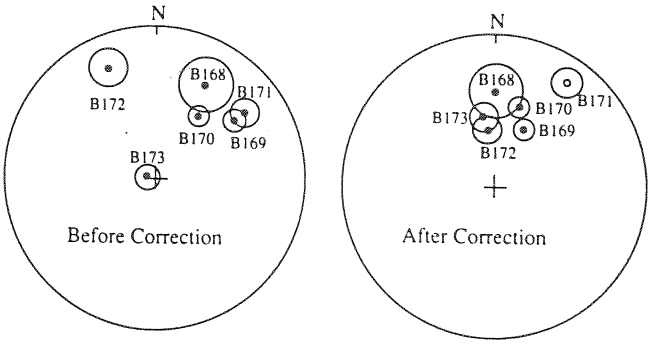


Fig. 8. Equal-angle nets of palaeomagnetic results, showing mean site vectors and α_{95} circles of confidence, before and after correction for bedding tilt. Filled circles are in the lower hemisphere, and open circle in the upper hemisphere.

probably carried by magnetite. Within sites the directional scatter of the high temperature component is low, with α_{95} values all <15° (Table 1). The two fold tests give ambiguous results: the first, between sites B170 and B171 on two limbs of the WNW-vergent fold in section E, shows a marginal, but insignificant improvement in grouping after the application of a bedding correction. A fracture cleavage is developed on the lower limb of this fold, and although there is no evidence for extensive recrystallization or the development of a pervasive fabric in thin section, we suggest that the site on the lower limb carries a magnetization acquired after folding. The second fold test is from the upright, horizontal syncline in the SE of the section (subsection H), with one site on each limb. Scatter is significantly reduced after the application of a bedding correction, suggesting that in this case the magnetization predates folding.

The mean directions from all the sites (Fig. 8), show a distinctive grouping after correction, except for that from site B171, on the overturned limb of the first fold. This suggests that, with the exception of this site, the palaeomagnetic samples carry a magnetisation which predates folding. The mean site direction, excluding this site is Dec. = 006° Inc. = 39°, $\alpha_{95} = 14°$, close to both the Late Cretaceous and Miocene reference directions for Iberia, indicating that there has been no measureable rotation of the section. This indicates that the kinematic data acquired in this section can be directly related to the overall tectonic framework of the Betics.

The palaeomagnetic results from the capas rojas (Campanian–Maastrichtian) limestones are all from the lower thurst sheet, the lowest tectonic unit. Excluding site B171, they give a mean declination anomaly of +8° ± 14°. This is consistent with the declination anomaly from the Upper Jurassic limestones of the Almirez block of Allerton *et al.* (1993) (−20° ± 22°), supporting the idea that the lower thrust sheet is part of the Almirez block, which has not suffered a significant cumulative rotation during the Miocene deformation.

In contrast, Allerton *et al.* (1993) obtain the largest magnitude rotations from Jurassic blocks within the dominantly Triassic upper thrust sheet at Carro (+190° ± 11°) and Gonzalo (+170° ± 12°). Internal deformation within this sheet can be assumed to be significant.

Palaeomagnetic results from the Baranco Salada section, being confined to the lower thurst sheet can only constrain the original orientations within this thrust sheet, and structures within the middle thrust sheet which post-date emplacement of the middle sheet over the lower sheet. The original orientations of structures within subsections A and B cannot be constrained using the palaeomagnetic results.

Limited data from the early thrust of the middle sheet over the lower sheet suggest that the middle sheet moved towards the east. The general overthrust direction appears to be dominantly towards the east, although a wide variation is seen, perhaps reflecting differential transport directions, or alternatively, rotation during overthrusting. This result is

Table 1. *Palaeomagnetic data from sites in the capas rojas red marly limestones of the Barranco Salada section*

Site	*Dec*	*Inc*	*k*	α_{95}	*N*	*n*	Dec	Inc	Dip	Direction
	Before correction						After correction			
B168	028	21	14	14	9	9	001	26	56	190
B169	053	24	85	6	8	8	027	43	42	285
B170	034	37	77	6	8	8	017	32	24	306
B171	055	18	60	7	9	9	034	−10	42	140
B172	336	14	39	9	8	8	354	48	60	112
B173	283	84	44	9	8	7	352	40	47	340

N = total number of samples; n = number of samples in statistics; α_{95} = radius of 95% circle of confidence about mean direction; k = dispersion parameter. Bedding is specified by dip and dip direction.

largely consistent with the more regional studies of Frizon de Lamotte *et al.* (1989).

Discussion

This section through the Subbetic can be interpreted as the result of a major overthrust of basinal sediments of the Median Subbetic over marginal shelf facies of the Internal Subbetic. Reconstruction of the section requires a minimum shortening from 8 to 2 km of section, although displacement on the thrust may be considerably more than this, particularly if Lower Cretaceous and Jurassic sediments overlie the Triassic. The dominant overthrusting direction appears to be roughly towards the E; in a back-thrust relationship to the main accretionary wedge, and strongly oblique to the Iberian margin, which has a general ESE trend. Strike-slip faults are relatively common, but show no dominant orientations. None appear to have a significant displacement, and often reactivate earlier steep dip-slip faults. We believe that these are related to Mid–Late Miocene to recent strike-slip on the Crevillente fault (both dextral and sinistral) and elsewhere in the Eastern Betics.

We consider that the Miocene orogen within the Subbetic zone commenced with thrust transport oblique to the Iberian margin, which constructed an accretionary wedge parallel to that margin. Rotation about vertical axes accompanied thrusting, reflecting the strong component of vorticity within the wedge. At some stage, the geometry of the wedge became critical, and further shortening within the wedge was arrested. Oblique convergence continued, resolved into a component of shortening orthogonal to the margin on a frontal thrust, i.e., the Subbetic thrust over the Prebetic; and a component of strike-slip within the wedge. Within such a framework, the lateral motion on the Crevillente fault would be transfered onto a basal detachment at depth, and not cut the basement of the Iberian Meseta.

Hermes (1978) relates the chaotic Triassic upper thrust sheet to tectonic fragmentation associated with strike-slip movement on the Crevillente fault zone, an idea that was developed by de Smet (1984). We suggest that this unit is not limited to the Central zone, but occurs as a shallow thrust sheet over much of the Subbetic, as we recognize this facies from Caravaca to south of Zarcilla de Ramos. Hermes (1978) and de Smet (1984) develop the idea that the variation in Jurassic stratigraphy within the central zone of the Subbetic is the result of tectonic juxtaposition within a major strike-slip fault system. De Smet (1984) recognizes the several situations where Lower Cretaceous sediments are in tectonic contact at their base with the Triassic. We submit that given the degree of internal deformation of the Triassic, such a tectonic contact would be difficult to distinguish from a direct stratigraphic contact, and that some of these apparently normal faults may be sites of direct deposition of Cretaceous over Triassic. We suggest that the Triassic, with its abundant evaporitic horizons, may have acted as a major crustal detachment during Mid–Late Jurassic rifting (Hermes 1978), and that Lower Liassic platform sediments were broken into isolated fault blocks during this event. Such an irregular basin configuration may be expected to produce very variable depositional environments; and a complex pattern of sedimentary facies and stratigraphic relationships during the rifting phase (Transitional stage, Hermes 1978).

Conclusions

The section can be divided into three tectonic units: an upper thrust sheet of brecciated Triassic sediments and intrusions; a middle thrust sheet of Cenomanian–Turonian marly limestones; and a lower thrust sheet of Jurassic oolitic limestone shelf facies unconformably overlain by Campanian-Palaeocene marls and marly limestones. The basinal sediments of the middle sheet were thrust over the lower; the combined sequence was folded and imbricated, then overthrust by the Triassic of the upper sheet. A footwall duplex of Cenomanian–Turonian sediments was transported with the upper sheet. This structural development can be related to complexities in the hanging wall and footwall of a single overthusting of the basinal sequence over the marginal shelf sediments.

Palaeomagnetic results show no significant rotation of the lower thrust sheet; a similar result to that obtained from the Almirez block of Allerton *et al.* (1993), with which it probably is in structural continuity. Kinematic results from the footwall can be directly related to a regional reference. Thrust transport directions are dominantly to the east, in a back-thrust relationship to the orogen as a whole, and strongly oblique to the Betic trend (ENE). Strike-slip faulting, which occurs throughout the section, is a relatively insignificant, late-stage feature.

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